

# Analysis of edX Courses: Certification Rates Across Subjects

## INTRODUCTION

**edX**, a platform developed by **MIT** and **Harvard**, provides access to **online courses** for millions of learners worldwide. However, not all courses achieve the same level of completion. This project investigates **how certification rates vary across different subjects** and institutions to better understand learner behavior and course effectiveness.



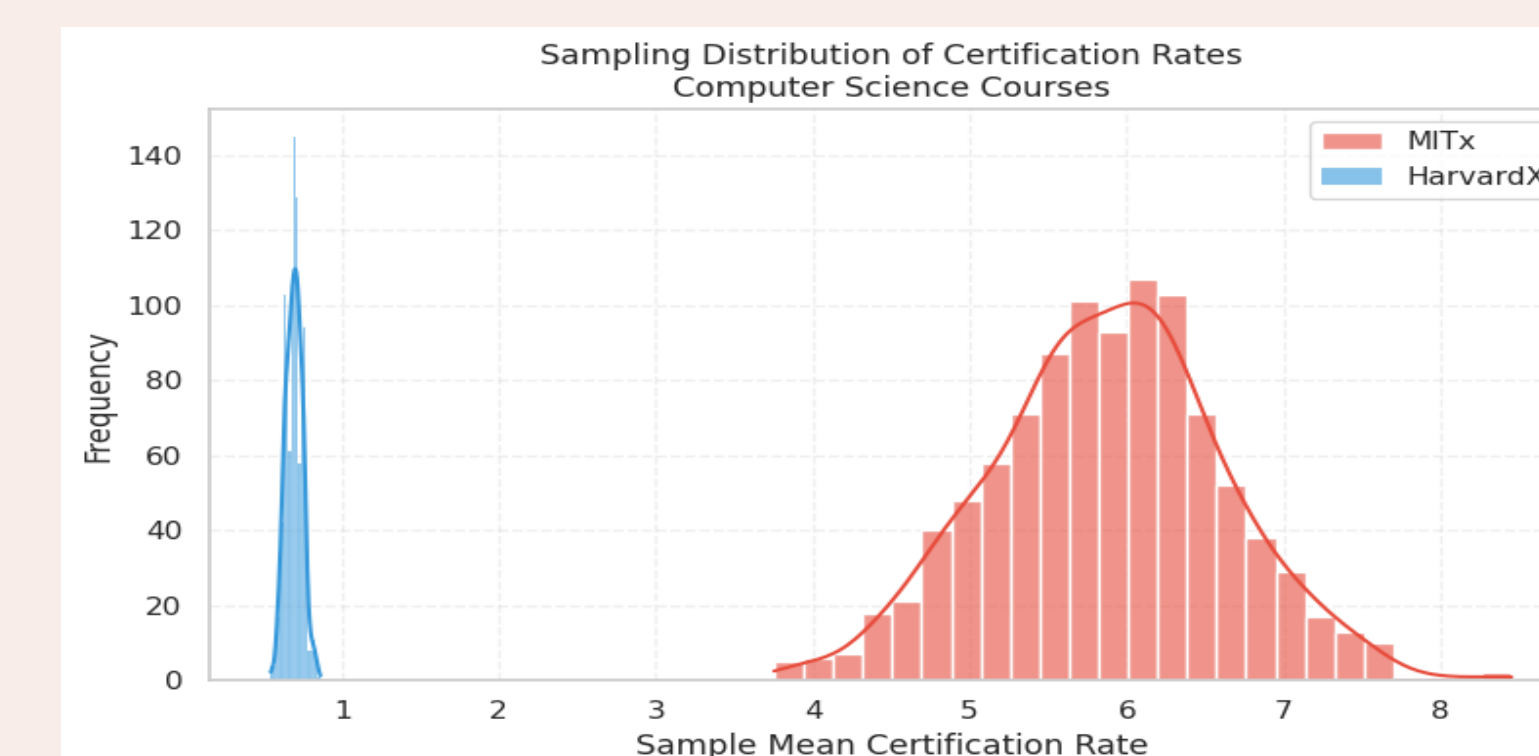
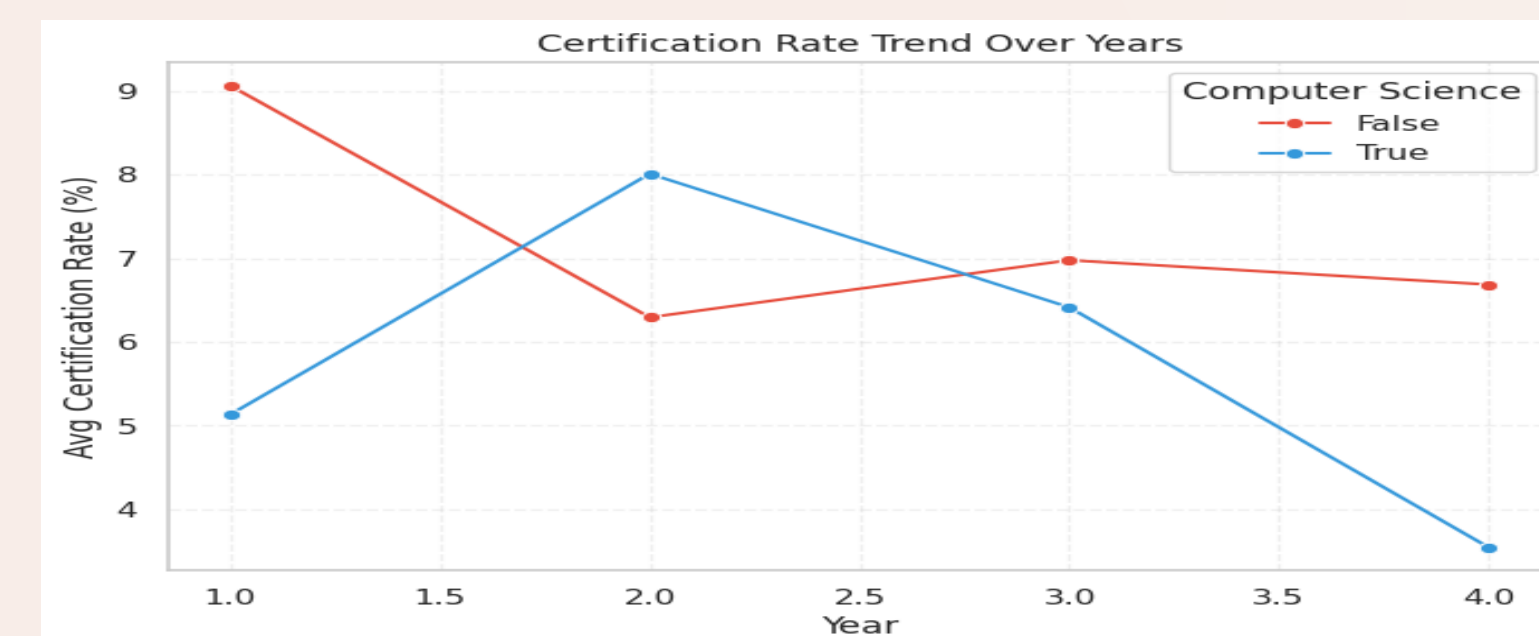
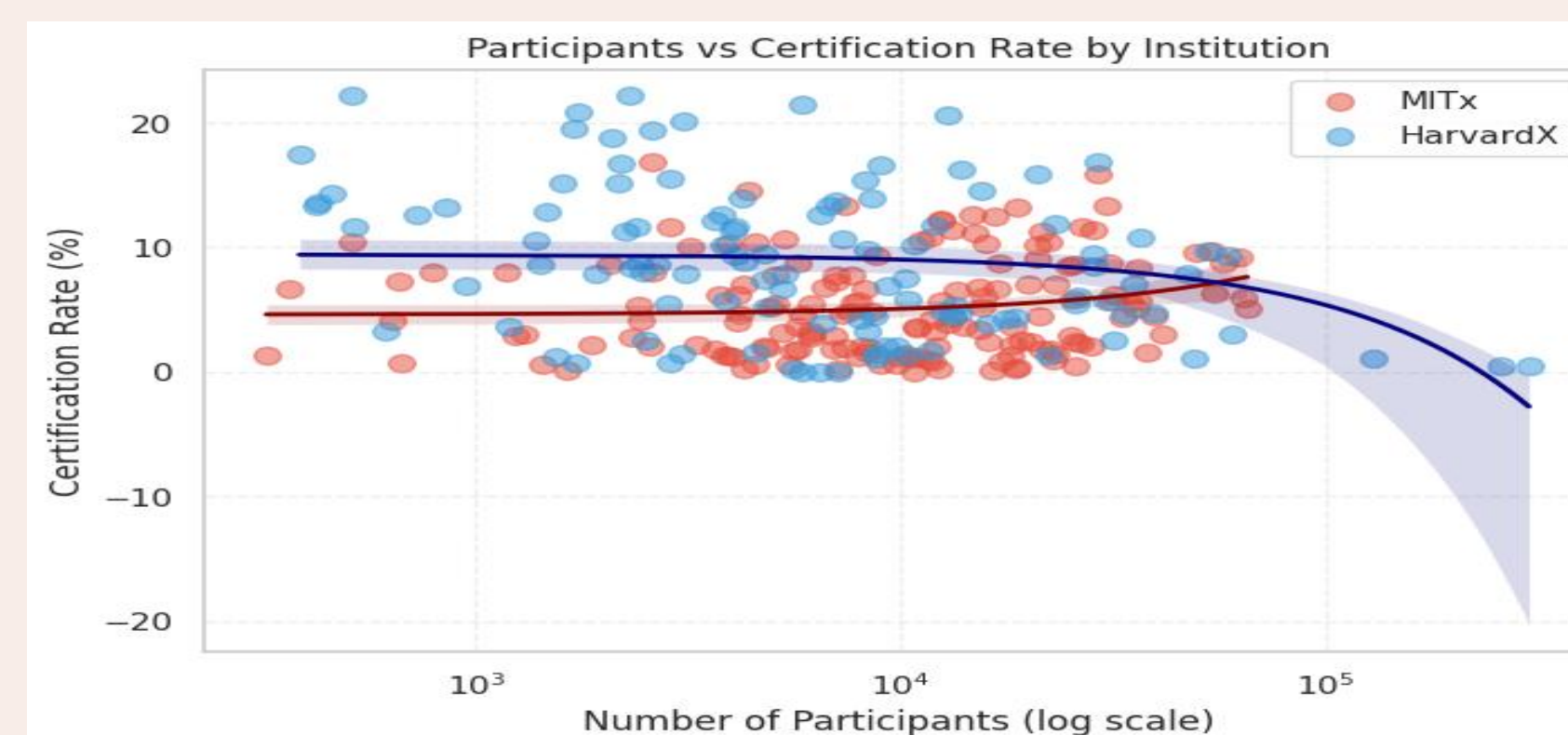
**Research Question:** Do certification rates differ significantly across different course subject areas?

### Why It Matters:

Identifying these differences helps improve course design, increase student engagement, and enhance overall learning outcomes.

## Analysis

Certification rates are **generally low across most courses**, with noticeable variation between subjects. **Some fields achieve higher completion rates**, while others remain lower, showing that subject area and course characteristics influence certification outcomes.



## Hypothesis

Certification rates in Computer Science courses may vary between institutions (MIT and Harvard).

- $H_0$ : No significant difference in certification rates between MIT and Harvard (Computer Science)
- $H_1$ : There is a significant difference in certification rates between MIT and Harvard courses in Computer Science.

## Hypothesis

Do certification rates differ significantly across different course subject areas?

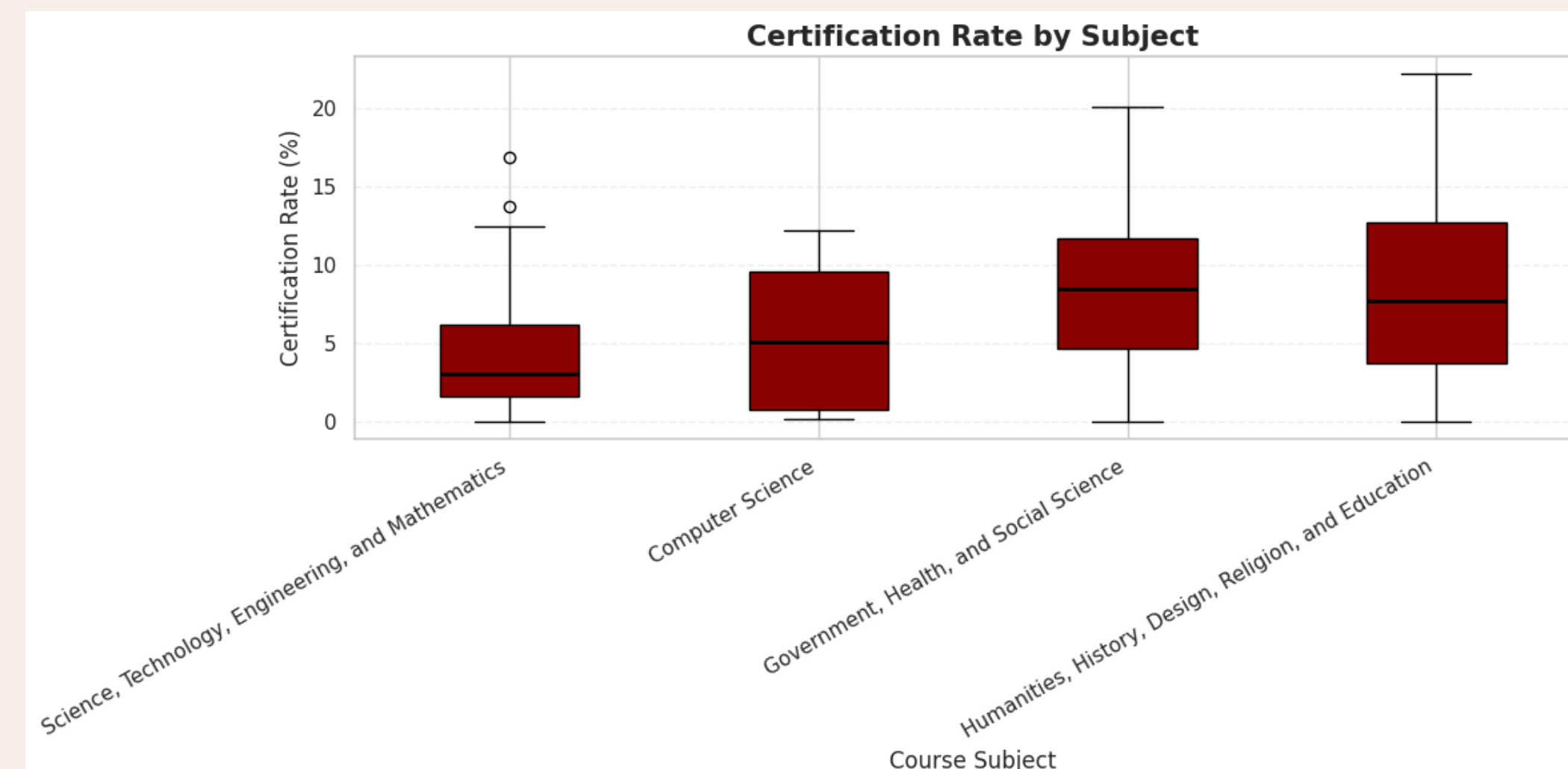
- $H_0$ : No significant difference in certification rates across different course subject areas
- $H_1$ : There is a certification rates differ significantly across subject areas .

## Methodology



### Data Source & Dataset Description:

- HarvardX & MITx dataset (2012–2016)
- 290 courses, 4.5M learners, 250K certifications
- Includes engagement and course data



## Conclusion

- Certification rates **differ significantly** across subjects.
- Some subjects achieve **higher completion rates** than others.
- MIT and Harvard **courses show differences in certification** outcomes.
- Most courses have **relatively low** completion rates.
- Visualizations show **variability in certification rates within subjects**.

### Implications:

- Improving **course structure** may **increase completion** rates
- **Higher learner engagement** can **improve certification** outcomes